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Wogonin induces mitochondrial apoptosis and synergizes with venetoclax in diffuse large B-cell lymphoma

Ye Lin [a,b], Xia Jiang [a,b,c], Mengting Zhao [b], Youhong Li [a,b,c], Lili Jin [a,b,c], Sumeng Xiang [a,b,c], Renzhi Pei [a,c], Ying Lu [a,c,**], Lei Jiang [a,b,*]

[a] Department of Hematology, The Affiliated People's Hospital of Ningbo University, Ningbo, China
[b] Department of Pathology, and Zhejiang Key Laboratory of Pathophysiology, School of Basic Medical Sciences, Health Science Center, Ningbo University, Ningbo, China
[c] Institute of Hematology, Ningbo University, Ningbo, China

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ABSTRACT

Diffuse large B-cell lymphoma (DLBCL) is among the most aggressive hematological malignancies and patients are commonly treated with combinatorial immunochemotherapies such as R-CHOP. Till now, the prognoses are still variable and unsatisfactory, depending on the molecular subtype and the treatment response. Developing effective and tolerable new agents is always urgently needed, and compounds from a natural source have gained increasing attentions. Wogonin is an active flavonoid extracted from the traditional Chinese herbal medicine *Scutellaria baicalensis* Georgi and has shown extensive antitumor potentials. However, the therapeutic effect of wogonin on DLBCL remains unknown. Here, we found that treatment with wogonin dose- and time-dependently reduced the viability in a panel of established DLBCL cell lines. The cytotoxicity of wogonin was mediated through apoptosis induction, along with the loss of mitochondrial membrane potential and the downregulation of BCL-2, MCL-1, and BCL-xL. In terms of the mechanism, wogonin inhibited the PI3K and MAPK pathways, as evidenced by the clear decline in the phosphorylation of AKT, GSK3β, S6, ERK, and P38. Furthermore, the combination of wogonin and the BCL-2 inhibitor venetoclax elicited synergistically enhanced killing effect on DLBCL cells regardless of their molecular subtypes. Finally, administration of wogonin significantly impeded the progression of the DLBCL tumor in a xenograft animal model without obvious side effects. Taken together, the present study suggests a promising potential of wogonin in the treatment of DLBCL patients either as monotherapy or an adjuvant for venetoclax-based combinations.

1. Introduction

Diffuse large B-cell lymphoma (DLBCL), the most frequent type of non-Hodgkin lymphoma (NHL), is a highly heterogeneous lymphoid neoplasm that arises *de novo* or transforms from other indolent lymphomas (Sehn and Salles, 2021; Susanibar-Adaniya and Barta, 2021). The cell-of-origin classification defines two major subgroups of DLBCL including the germinal center B-cell-like (GCB) and the activated B-cell-like (ABC) subtypes (Alizadeh et al., 2000), differ not only in oncogenic addictions but also in clinical outcomes (Rosenwald et al., 2002; Schmitz et al., 2018). The GCB DLBCL is characterized by the existence of chromosomal translocations of BCL-2 and gain-of-function mutations of EZH2, while constitutive activation of nuclear factor kappa B (NF-κB) by BCR signaling is the hallmark of the ABC DLBCL (Pasqualucci and Dalla-Favera, 2018). In addition, primary mediastinal B-cell lymphoma is another subtype of DLBCL and comprises about 5 % of all NHL (Roschewski et al., 2020). Despite the emergency of novel agents such as antibody drug conjugates, bispecific antibodies, small molecular inhibitors, as well as CAR T-cells (Cheson et al., 2021; Frontzek et al., 2022), chemo-immunotherapy with R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine and prednisone) remains the standard frontline treatment for most newly diagnosed DLBCL patients

Abbreviations: DLBCL, Diffuse large B-cell lymphoma; NHL, Non-Hodgkin lymphoma; GCB, Germinal center B-cell-like; ABC, Activated B-cell-like; R-CHOP, Rituximab, cyclophosphamide, doxorubicin, vincristine and prednisone; MMP, Mitochondrial membrane potential; CI, Combination index; PI, Propidium iodide; PBS, Phosphate-buffered saline; BCA, Bicinchoninic acid; H&E, Hematoxylin and eosin.

* Correspondence to: Lei Jiang, Department of Pathology, School of Basic Medical Sciences, Health Science Center, Ningbo University, 315211 Ningbo, China.
** Correspondence to: Ying Lu, Department of Hematology, The Affiliated People's Hospital of Ningbo University, 315101 Ningbo, China.
E-mail addresses: rmluying@nbu.edu.cn (Y. Lu), jianglei@nbu.edu.cn (L. Jiang).

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irrespective of molecular profiles, resulting in a more favorable prognosis in patients with the GCB subtype (Lenz et al., 2008). However, approximately 30–40 % of patients will relapse after a period of remission (Crump et al., 2017). The majority of those patients cannot be cured, highlighting an unmet medical need to develop new therapies.

Wogonin is one of the active flavonoids extracted from the root of *Scutellaria baicalensis* Georgi, a traditional Chinese herbal medicine that has been applied in the treatment of hypertension, diarrhea, insomnia, respiratory infections and other diseases for thousands of years and is now officially listed in Chinese Pharmacopoeia (Zhao et al., 2016; Zhao et al., 2019). Wogonin possesses diverse biological activities (Huynh et al., 2020), with the most attractive being the antitumor effect which is mediated through different mechanisms against various cancer types (Huynh et al., 2017; Banik et al., 2022). Treatment with wogonin evokes apoptosis in human cell lines of breast cancer (Chung et al., 2008; Huang et al., 2012), colorectal cancer (Kim et al., 2012), prostate cancer (Sun et al., 2022), neuroblastoma (Ge et al., 2015), nasopharyngeal carcinoma (Chow et al., 2011), and osteosarcoma (Lin et al., 2011), and triggers ferroptosis in pancreatic cancer cells (Liu et al., 2023). Two independent studies report that wogonin arrests cell cycle at G1 phase in colorectal carcinoma (He et al., 2013) and hepatocellular carcinoma cells (Hong et al., 2020). Wogonin also restrains tumor growth by targeting angiogenesis and lymphangiogenesis (Kimura and Sumiyoshi, 2013; Song et al., 2013). More interestingly, wogonin plays a bidirectional role in regulating the complex interaction between inflammation and cancer, as demonstrated by evidences that wogonin not only suppresses the melanoma-associated inflammation (Li et al., 2021), but also reduces the incidence and inhibits the progression of colitis-initiated colorectal adenomas in azoxymethane/dextran sulfate sodium-induced mice (Yao et al., 2014). In addition to the killing effect, wogonin is capable of inducing terminal differentiation of NB4 and U937 leukemia cells and thus prevents the uncontrollable proliferation (Zhang et al., 2008a; Zhang et al., 2008b). In the case of NHL, wogonin exhibits cytotoxicity in a panel of mantle cell lymphoma cell lines (Mino, Jeko-1 and REC-1) (Xu et al., 2018) and a Burkitt's lymphoma cell line (Raji) (Wu et al., 2017), indicating the antilymphoma potential. However, to the best of our knowledge, information regarding the effect of wogonin on DLBCL remains scarce.

The current study showed that treatment with wogonin reduced the viability in a panel of established DLBCL cell lines by inducing apoptotic cell death, accompanied by the loss of mitochondrial membrane potential (MMP) and the downregulation of antiapoptotic members of the BCL-2 family of proteins. Moreover, wogonin inhibited the activation of PI3K and MAPK pathways, and potentiated the killing effect of the BCL-2 inhibitor venetoclax in DLBCL cells. The anti-DLBCL activity of wogonin was also confirmed in a SUDHL-4 cell-derived xenograft mouse model. Our data thus demonstrates a great potential of wogonin in treating patients with DLBCL.

2. Materials and methods

2.1. Drugs

Wogonin (#HY-N0400, purity 99.92 %) was purchased from MedChemExpress (NJ, USA). Venetoclax (#S8048) was purchased from Selleck Chemicals (Houston, TX, USA).

2.2. Cell cultures

Human DLBCL cell lines including SUDHL-4, SUDHL-16, OCI-Ly3, OCI-Ly10, DB, Farage, Pfeiffer, and U2932 were purchased from American Type Culture Collection (ATCC, Manassas, VA, USA) or Deutsche Sammlung von Mikroorganismen und Zellkulturen (DSMZ, Braunschweig, Germany). All cells were cultured in Roswell Park Memorial Institute 1640 (RPMI-1640) medium (Hyclone, Utah, USA) supplemented with 10 % fetal bovine serum (AusgeneX, Molendinar, QLD, Australia), 100 U/mL penicillin, 100 U/mL streptomycin, and 2 mM L-glutamine (all Solarbio, Beijing, China) at 37 °C in a humidified atmosphere of 5 % CO_2.

2.3. Cell viability

DLBCL cells were seeded in 96-well plates (5000–30,000 cells in 100 μL per well, depending on the cell type) in triplicates and treated with vehicle (0.1 % DMSO) or various concentrations of wogonin (1, 5, 10, 50, and 100 μM). After 24 and 48 h of treatment, 10 μL of MTS (Promega, Fitchburg, WI, USA) was added to each well and incubated for 4 h. The absorbance at 490 nm was measured using a microplate reader (Molecular Devices, San Jose, CA, USA). For two-drug combinations, cells were treated with increasing concentrations of venetoclax (10^{-5}–10^{-1} μM) in the absence or presence of wogonin for 24 h. According to the method of Chou-Talalay (Chou, 2010), combination index (CI) values were calculated and Fa-CI plots were generated utilizing CompuSyn software (version 1.0). Synergy of the two drugs is defined as a CI less than 1.

2.4. Apoptosis assay

Cell apoptosis was determined using an Annexin V/propidium iodide (PI) dual staining kit (#V13241, Invitrogen, Shanghai, China) according to the manufacturer's instructions. After treatment with or without wogonin for 24 h in 6-well plates, DLBCL cells were harvested, washed thrice with cold phosphate-buffered saline (PBS), and resuspended in 100 μL 1 × Annexin-binding buffer. Subsequently, cells were stained with 5 μL fluorescein isothiocyanate-conjugated Annexin V and 1 μL PI (100 μg/mL) at room temperature for 15 min protected from light. Once another 400 μL 1 × Annexin-binding buffer was added, samples were subjected to flow cytometry (CytoFLEX S, Beckman Coulter Life Sciences, Indianapolis, IN, USA) immediately. Data were analyzed using the CytExpert software (Beckman Coulter, version 2.4) and Annexin V-positive cells were designated as apoptotic cells.

2.5. MMP detection

MMP alterations were detected using the JC-1 staining kit (#C2003S, Beyotime, Hangzhou, China) according to the manufacturer's instructions. Briefly, DLBCL cells were seeded in 6-well plates and treated with or without wogonin for 24 h. Cells were collected, washed with PBS twice, resuspended in culture medium, and incubated with pre-prepared JC-1 working solution at 37 °C in a 5 % CO_2 incubator for 20 min. After washing with staining buffer twice, all samples were subsequently analyzed using flow cytometry. Results were presented as the percentage of cells with MMP loss.

2.6. Western blot

Cells were harvested, washed with PBS and lysed in cold Cell Lysis Buffer (#9803, Cell Signaling Technology) supplemented with protease and phosphatase inhibitors (Roche, Mannheim, Germany) on ice for 30 min. The supernatant was collected after centrifugation (12,000 rpm, 4 °C, 10 min) and protein concentration was quantified using a bicinchoninic acid (BCA) kit (TransGen Biotech, Beijing, China) according to the manufacturer's instructions. Blue Loading Buffer containing dithiothreitol (#7722, CST) was added to each sample, followed by denaturation at 95 °C for 5 min. Equal amounts of proteins (25 μg per lane) were electrophoresed on sodium dodecyl sulfate (SDS)-polyacrylamide gels and transferred to polyvinylidene difluoride (PVDF) membranes (Bio-Rad Laboratories, Hercules, CA). The transblotted membranes were blocked in blocking buffer (#P30500, New Cell & Molecular Biotech Co., Ltd., Suzhou, Jiangsu, China) for 20 min, washed with double distilled water, and coated with specific primary antibodies at 4 °C overnight. After washing with tris-buffered saline containing 0.1 %

Tween 20 (TBST) three times (15 min each), membranes were incubated with the corresponding secondary antibodies, goat anti-rabbit IRDye 680RD and goat anti-mouse IRDye 800CW (LI-COR Biosciences, Lincoln, NE, USA), in the dark for 1 h at room temperature. The membranes were then washed three times again, and scanned using an Odyssey infrared imaging system (LI-COR Biosciences). For proteins with similar molecular weights, the membranes were incubated with stripping buffer (#P0025N, Beyotime) for 5 min, washed three times with TBST, and then re-probed after blocking. Densitometry measurements were carried out using Image J software. Primary antibodies used in the current study included PARP (#9532; CST, 1:1000); MCL-1 (#16225–1-AP; Proteintech, 1:1000); BCL-xL (#10783–1-AP; Proteintech, 1:2000); BCL-2 (#4223; CST, 1:1000); AKT (#2920; CST, 1:1000); phospho-AKT (Ser473; #4060; CST, 1:1000); glycogen synthase kinase-3β (GSK3β; #9315; CST, 1:1000); phospho-GSK3β (Ser9; #9322; CST, 1:1000); ribosomal protein S6 (RPS6; #2217; CST, 1:1000); phospho-RPS6 (Ser235/236; #4858; CST, 1:1000); ERK (#4696; CST, 1:1000); phospho-ERK (Thr202/Tyr204; #4370; CST, 1:1000); P38 (#9212; CST, 1:1000); phospho-P38 (Thr180/Tyr182; #9216; CST, 1:1000); and GAPDH (#60004–1-Ig; Proteintech, 1:10000).

2.7. Xenograft study

All the animal experiments were approved by the Animal Care Committee of Ningbo University Health Science Center and were conducted in accordance with guidelines. Five-week-old female BALB/c-Nude mice were purchased from GemPharmatech (Nanjing, China) and were housed under a pathogen-free condition. For the establishment of DLBCL xenograft model, SUDHL-4 cells (1×10^6) were subcutaneously injected in the right flank of the mice. Once a lump formed and the size reached approximately 100 mm^3, the mice were randomly assigned to two groups ($n = 3$ for each group) and were treated intraperitoneally with or without wogonin at 25 mg/kg every other day for two weeks. Mice were monitored and the body weight was recorded throughout the study. Tumor volume was measured with a digital caliper and was calculated using the following formula: $V = (\text{length} \times \text{width}^2)/2$. At the end of the experiment, all tumors were excised, photographed, weighed, and fixed in formalin for histological examination.

2.8. Histopathological analysis

Hematoxylin and eosin (H&E) staining was performed according to standard protocols on 4-μm-thick sections prepared from paraffin-embedded blocks using a microtome (LEICA RM2245, Germany). Mitotic figures were counted in 10 high-power fields of each H&E-stained slide under a light microscope (LEICA DM 2500, Germany). The proliferative activity of tumors was assessed by immunohistochemical staining using the Ki-67 antibody (LabVision, 1:1000) as previously described (Zhao et al., 2023).

2.9. Statistical analysis

All results were presented as mean ± standard deviation (SD) of at least three independent experiments. Differences between groups were determined by Student's *t*-test using Prism software package (GraphPad Software Inc., San Diego, CA, USA). A p value less than 0.05 was considered statistically significant.

3. Results

3.1. Wogonin effectively decreases the survival of DLBCL cells in vitro

The chemical structure of wogonin is shown in Fig. 1A. The *in vitro* anti-DLBCL property of wogonin was firstly assessed in multiple cell lines including GCB (SUDHL-4, SUDHL-16, DB, Farage, and Pfeiffer) and ABC (OCI-Ly3, OCI-Ly10, and U2932) subtypes. After exposure to wogonin at 1, 5, 10, 50, and 100 μM for 24 and 48 h, the viability of these cell lines was determined *via* the MTS assay. Even inhibitory effects of wogonin used from 1 to 10 μM were not apparent and varied among different cell lines, higher concentrations of wogonin (50 and 100 μM) strikingly reduced the viability in a time-dependent manner in each cell line (Fig. 1B-E and Supplementary Fig. S1). Typically, in SUDHL-4 cells (Fig. 1B), treatment with wogonin at 50 and 100 μM reduced the viable rate to 54 % and 8 % at 24 h, and further to 12 % and 3 % at 48 h, respectively. Calculated IC_{50} values of wogonin at 24 and 48 h were listed in Table 1, showing divergent responses to wogonin in these DLBCL cell lines. These results demonstrated that wogonin effectively decreased the survival of DLBCL cells with distinct molecular backgrounds *in vitro*.

3.2. Wogonin induces mitochondria-dependent apoptosis in DLBCL cells

Next, we sought to investigate whether wogonin could induce apoptosis in DLBCL cells using Annexin V/PI dual staining by flow cytometry. Compared to the control, treatment with wogonin at 50 and 100 μM for 24 h increased the percentage of Annexin V-positive cells from 5 % to 14 % and 61 %, respectively, in SUDHL-4 cells (Fig. 2A), indicating that wogonin triggered apoptosis in a dose-dependent manner which was in line with the cell viability data. Considering the importance of mitochondria in the process of apoptosis, we subsequently determined the alterations in MMP in response to wogonin through mitochondria-selective JC-1 staining. JC-1 forms aggregates in normal mitochondria and emits red fluorescence but becomes monomers in depolarized mitochondria and emits green fluorescence. As shown in Fig. 2B, the number of cells with MMP loss gradually elevated in wogonin-treated SUDHL-4 cells as the concentration escalated. The apoptotic event is regulated by various molecules, among which the BCL-2 family proteins are the most critical. Western blot analysis revealed that wogonin treatment led to a prominent reduction in the level of antiapoptotic proteins including BCL-2, MCL-1, and BCL-xL (Fig. 2C). Meanwhile, a significant cleavage of PARP was observed in the presence of wogonin (Fig. 2C), confirming the apoptosis-inducing ability of wogonin. These results demonstrated that wogonin induced mitochondrial apoptosis by regulating antiapoptotic members of the BCL-2 family in DLBCL cells.

3.3. Wogonin inhibits the activation of the PI3K and MAPK signaling

Since the hyperactivation of the PI3K and MAPK signaling frequently occurs in DLBCL pathogenesis, we then examined the phosphorylation status of several critical molecules that were involved in these pathways to unravel the potential mechanism underlying wogonin-induced lethality. After treatment with wogonin for 24 h, it was observed that AKT phosphorylation at Ser473 site in SUDHL-4 cells was almost abolished, even though the total amount of AKT was decreased by a lesser extent (Fig. 3A). Furthermore, the phosphorylation of GSK3β at Ser9 and of S6 at Ser235/236, two direct downstream substrates of AKT, was also dramatically inhibited by wogonin (Fig. 3A). In regard to the MAPK signaling, wogonin treatment led to a substantial reduction in the level of ERK and P38 phosphorylation, at Thr202/Tyr204 and Thr180/Tyr182, respectively (Fig. 3B). These findings suggested that wogonin simultaneously regulated the PI3K and MAPK pathways in DLBCL cells.

3.4. Wogonin synergizes with the BCL-2 inhibitor venetoclax

To date, venetoclax is the only BCL-2 inhibitor that has been approved by the Food and Drug Administration (FDA) for treating patients with chronic lymphocytic leukemia (CLL). Unfortunately, venetoclax shows very limited activity in DLBCL patients, of which many overexpress the BCL-2 protein, when used as a single agent. The fact that wogonin downmodulates MCL-1 and BCL-xL prompts us to assess whether wogonin can enhance the sensitivity of DLBCL cells to

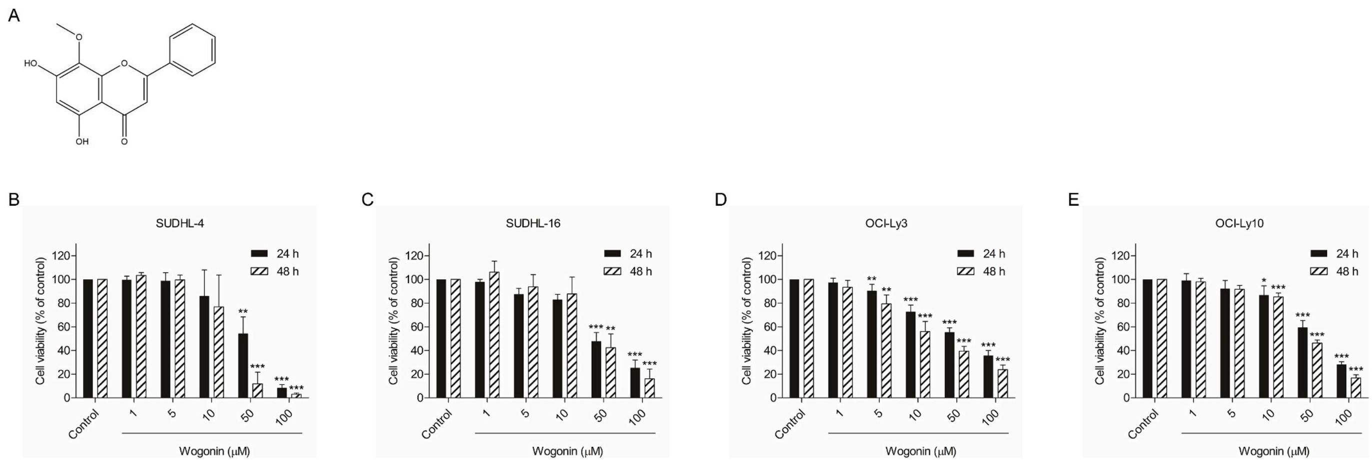

Fig. 1. Wogonin decreases the viability in DLBCL cells.
(A) The chemical structure of wogonin. SUDHL-4 (B), SUDHL-16 (C), OCI-Ly3 (D), and OCI-Ly10 (E) cells were treated with or without various concentrations of wogonin (1, 5, 10, 50, and 100 μM) for 24 and 48 h, followed by cell viability determination through the MTS assay. Data were shown as mean ± SD from at least three independent experiments. $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$ *versus* the control group at corresponding time points.

Table 1
Calculated IC_{50} for DLBCL cell lines.

Cell Line	IC_{50} (μM) 24 h	48 h	Subtype
SUDHL-4	52.19	18.64	GCB
SUDHL-16	41.42	38.38	GCB
OCI-Ly3	53.33	21.95	ABC
OCI-Ly10	56.22	38.90	ABC
DB	85.73	67.53	GCB
Farage	69.87	23.10	GCB
Pfeiffer	67.64	29.88	GCB
U2932	59.94	47.83	ABC

venetoclax. SUDHL-4 cells were sensitive to venetoclax monotherapy, as evidenced by the impaired viability in response to a wide range of concentrations. The combination of wogonin and venetoclax was significantly more effective than each drug alone (Fig. 4A, left). To clearly illustrate the cooperative drug interaction, the CI for each dose pair was calculated according to the Chou-Talalay method, generating values in the range of 0.11 to 0.95 (Supplementary Table S1), which was suggestive of synergy between wogonin and venetoclax (Fig. 4A, right). Unsurprisingly, a similar pattern was seen in SUDHL-16 cells (Fig. 4B and Supplementary Table S1). Notably, in venetoclax-resistant OCI-Ly3 and OCI-Ly10 cells, the addition of wogonin to venetoclax still produced synergistic killing effects, with CI values for most of combinations below 1 (Fig. 4C-D, and Supplementary Table S1). Thus, these data indicated that wogonin synergized with venetocalx in DLBCL cells, even in the difficult-to-treat ABC subtype.

3.5. Wogonin inhibits DLBCL tumor growth in SUDHL-4 xenografts

To validate the therapeutic effect of wogonin *in vivo*, a xenograft model was established by subcutaneous inoculation of SUDHL-4 cells in BALB/c-Nude mice. Administration of wogonin resulted in a prominent retardation in tumor growth, as shown by the average volume of tumors increased from 133 mm^3 to 2481 mm^3 in the control group and from 116 mm^3 to 534 mm^3 in the wogonin-treated group (Fig. 5A). Accordingly, the tumor weight was significantly reduced by wogonin treatment (Fig. 5B-C). During the treatment period, no apparent body weight loss was observed in mice, suggesting the excellent tolerance of wogonin (Fig. 5D). Moreover, tumor tissues were sectioned and subjected to histopathological examination. H&E staining and immunohistochemistry respectively showed a marked decrease in the frequency of mitotic figures (Fig. 5E) and Ki-67 positive cells (Fig. 5F) in wogonin-treated tumor sections, when compared with the controls. Collectively, these results demonstrated that wogonin treatment was able to suppress the progression of DLBCL tumor *in vivo*.

4. Discussion

DLBCL is still an incurable malignancy with limited treatment options. In past few years, compounds from natural plants with anti-lymphoma potentials are of great interest. To our knowledge, the current study demonstrates for the first time that wogonin exhibits the inhibitory activity against DLBCL both *in vitro* and *in vivo*, thus adding a new attribute on this flavonoid compound. Although we do not observe significant body weight loss in wogonin-treated mice, it is necessary to understand possible side effects caused by an overdose or a prolonged exposure before moving it into clinical trials. Qi *et al.* (Qi et al., 2009) have evaluated the acute and sub-chronic toxicities of wogonin in mice and Sprague-Dawley rats, respectively, and documented the LD_{50} of 286.15 mg/kg and a non-toxic dose of 30 mg/kg when administered *via* intravenous injection. Rats receiving wogonin at 60 and 120 mg/kg once daily show symptoms such as somasthenia, tachypnoea, and blepharon reclinatio, and the decrease in red blood cell and platelet counts as well as the increase in creatine phosphokinase occur 45 days after treatment. Once the treatment period prolongs to 90 days, 120 mg/kg wogonin stimulates heart injury in rats. Similar experiments are also performed in Beagle dogs and results show that intravenous infusion of wogonin at dosages of 15, 30, and 60 mg/kg per day for 90 days does not cause notable pathological alterations in major organs except for some adverse reactions including emesia, hyperptyalism, somatasthenia, and swollen snout (Peng et al., 2009). In addition to mature animals, wogonin treatment at 40 mg/kg induces developmental toxicities on pregnant rats and fetus as indicated by the reduction in both maternal body weight and birth weight (Zhao et al., 2011). These *in vivo* toxicological studies are beneficial for the design of a safe dosage and reasonable therapeutic duration when wogonin is applied in a clinical setting.

Apart from the well-known anticancer property, wogonin also shows benefits in counteracting inflammatory pathology. Treatment with wogonin is able to alleviate airway inflammation and relieve airway hyperresponsiveness in mice by inducing eosinophil or neutrophil apoptosis (Lucas et al., 2015; Bai et al., 2022). In the mouse model of dextran sulfate sodium-induced colitis, wogonin administration activates PPARγ to regulate colonocyte metabolism, reduces the abnormal expansion of Enterobacteriaceae, and ameliorates the intestinal

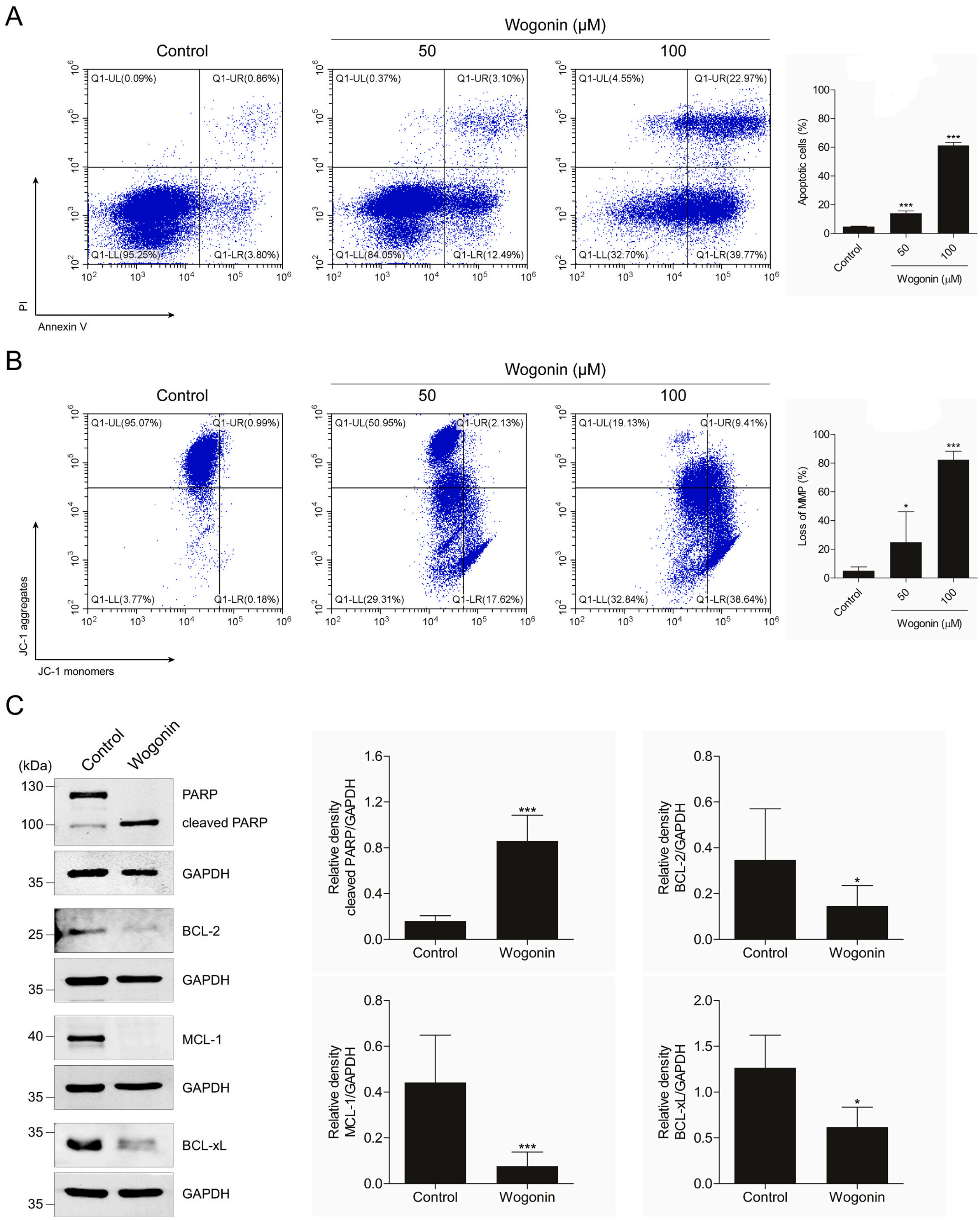

Fig. 2. Wogonin induces mitochondria-dependent apoptosis in DLBCL cells.
(A) SUDHL-4 cells were treated with vehicle or wogonin (50 and 100 μM) for 24 h and cellular apoptosis was detected through flow cytometry. Representative plots from three independent experiments were shown on the left side. The percentage of apoptotic (Annexin V-positive) cells was calculated and was shown on the right side. ****p* < 0.001 *versus* the control group. (B) SUDHL-4 cells were treated with vehicle or wogonin (50 and 100 μM) for 24 h, followed by staining with JC-1 dye. Representative flow cytometry plots from three independent experiments were shown and the percentage of cells with MMP loss was quantified. **p* < 0.05 and ****p* < 0.001 *versus* the control group. (C) Western blot analysis of PARP (full length and cleaved), BCL-2, MCL-1, and BCL-xL protein levels in SUDHL-4 cells treated with wogonin at 100 μM for 24 h. GAPDH was served as a loading control. Representative blots from three independent experiments and densitometric analysis were shown. **p* < 0.05 and ****p* < 0.001 *versus* the control group.

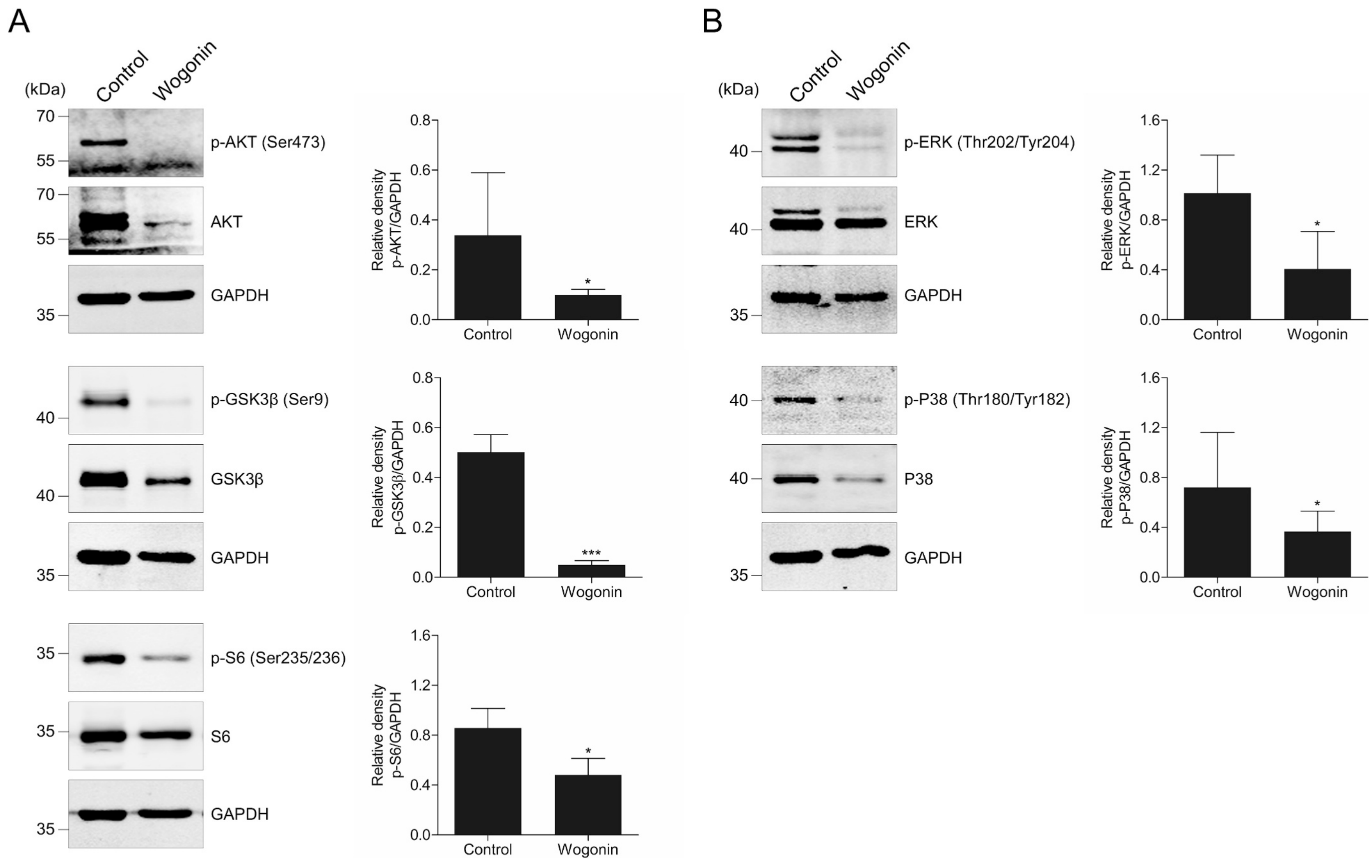

Fig. 3. Wogonin regulates the PI3K and MAPK signaling.
(A) SUDHL-4 cells were treated with vehicle or wogonin (100 μM) for 24 h. Levels of total and phosphorylated AKT, GSK3β, and S6 were analyzed by western blot. GAPDH was served as a loading control. Representative blots from three independent experiments and densitometric analysis were shown. $^{*}p < 0.05$ and $^{***}p < 0.001$ *versus* the control group. (B) SUDHL-4 cells were treated with vehicle or wogonin (100 μM) for 24 h. Levels of total and phosphorylated ERK and P38 were analyzed by western blot. GAPDH was served as a loading control. Representative blots from three independent experiments and densitometric analysis were shown. $^{*}p < 0.05$ *versus* the control group.

inflammation (Su et al., 2023). Li *et al.* (Li et al., 2017) suggest that wogonin effectively inhibits the inflammatory response and protects liver injury in EtOH-fed mice with alcoholic liver disease. Moreover, wogonin can attenuate mammary tissue damage, inflammation, and oxidative stress in lipopolysaccharide-induced murine mastitis (He et al., 2023). The antiviral and antifungal activities of wogonin are also documented in previous studies (Pal et al., 2003; Guo et al., 2007; Seong et al., 2018). Given the close association between chronic inflammation and DLBCL progression, the effect of wogonin on the tumor microenvironment of DLBCL is worth being investigated. Using different brain injury rat models, wogonin shows its potent neuroprotective effect *via* the prevention of the death of hippocampal neurons as well as the inhibition of the inflammatory activation of microglia (Lee et al., 2003), implying the potential application of wogonin in treating neurodegenerative disorders such as Alzheimer's disease (Zhu and Wang, 2015). In addition, wogonin acts as an anxiolytic with high affinity for the benzodiazepine site on the γ-aminobutyric acid_A receptor complex (Hui et al., 2002). Therefore, abovementioned findings including ours emphasize the therapeutic potential of wogonin for the treatment of both malignant and nonmalignant diseases in human.

The PI3K/AKT pathway is one of vital oncogenic signaling cascades and is frequently overactivated in lymphomagenesis. DLBCL patients with a high level of phosphorylated AKT experience a more rapidly exacerbating clinical course and a shortened survival time (Hasselblom et al., 2010; Hong et al., 2014). Even though this pathway is regarded as a hot therapeutic target, PI3K inhibitors, either approved for clinical use (idelalisib, copanlisib and duvelisib) or under clinical evaluation (buparlisib and umbralisib), do not bring inspiring benefits in DLBCL patients when used as monotherapy (Xu et al., 2021). Like AKT, ERK is highly expressed in DLBCL specimens particularly in the GCB subtype (Dai et al., 2011), and P38 hyperactivation is associated with poor response to CHOP therapy and inferior prognosis (Vega et al., 2015). More importantly, the concurrent activation of AKT, ERK and P38 is observed in the case of DLBCL, highlighting the aberrant activation of the MAPK pathway as well as a close crosstalk between PI3K and MAPK in DLBCL (Ogasawara et al., 2003). Hence, resistance to the PI3K inhibition is likely ascribed to the alternative stimulation of the MAPK signaling, endorsing a solid rationale for simultaneously targeting both pathways (Cao et al., 2019). Compared to the single-agent treatment, the combination of PI3K inhibitors with MAPK inhibitors yields encouraging efficacy in preclinical models (Gaudio et al., 2016), but is accompanied by unexpected adverse effects in clinical practice (Cao et al., 2019). ONC201, a small molecule that dually inhibits AKT and ERK, induces cell death in multiple cancer cell lines including the DLBCL type and shows acceptable tolerability in clinical tests of patients with solid tumors, preserving the feasibility of co-targeting PI3K and MAPK in DLBCL therapy (Prabhu et al., 2018; Cao et al., 2019). In our hands, we find that treatment with wogonin significantly inhibits the phosphorylation of AKT, ERK, and P38 in DLBCL cells and elicits a promising anti-DLBCL activity *in vitro* and *in vivo*. Although how wogonin regulates PI3K and MAPK pathways remains unclear, our data indeed indicate that wogonin can be applied as a candidate drug for the treatment of PI3K- and MAPK-dependent cancers including but not limited to DLBCL, and that components with PI3K and MAPK dual-targeting property from

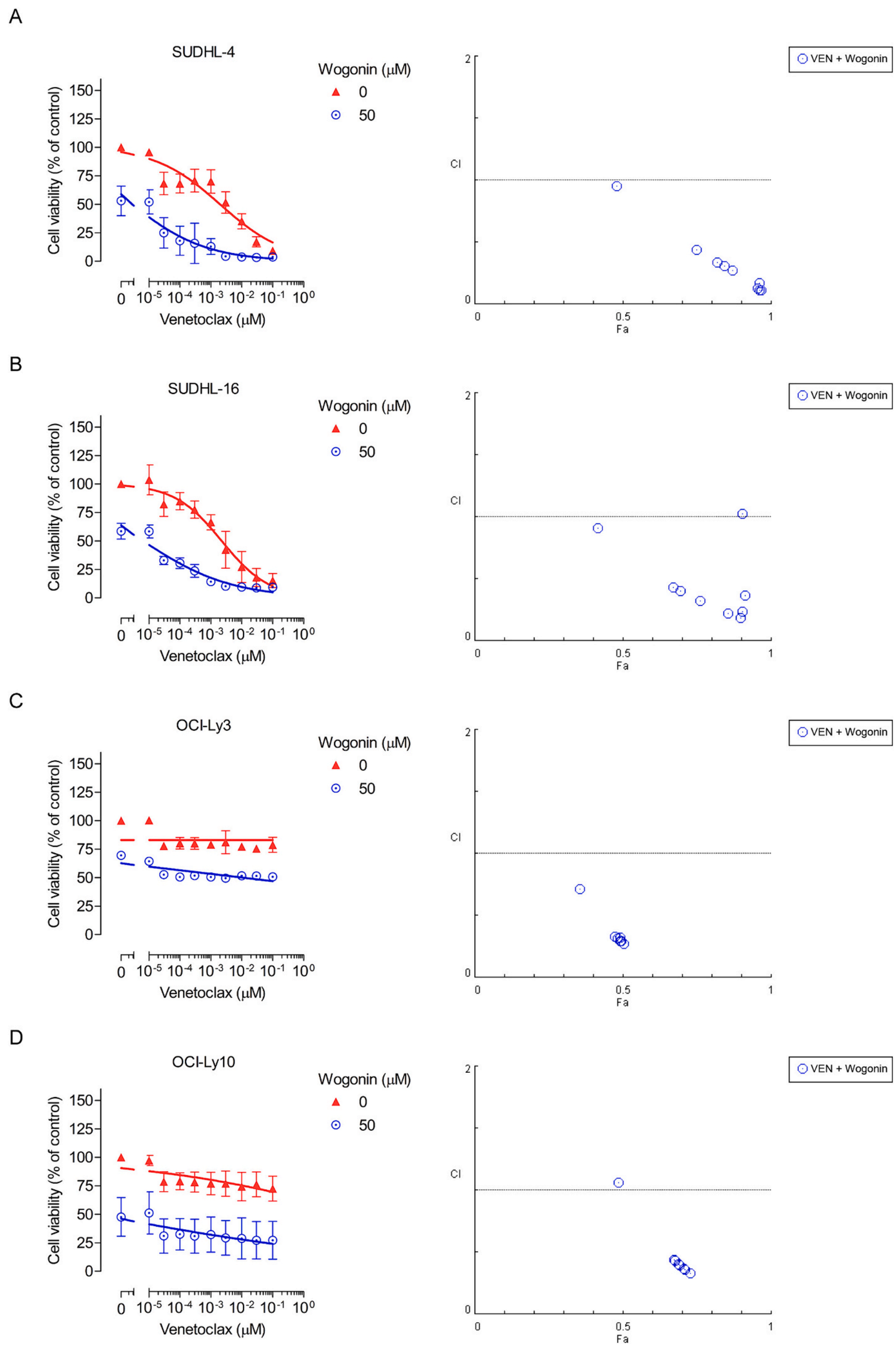

Fig. 4. Wogonin synergizes with venetoclax in DLBCL cells.
SUDHL-4 (A), SUDHL-16 (B), OCI-Ly3 (C), and OCI-Ly10 (D) cells were treated with venetoclax and wogonin at indicated concentrations, either alone or in combination, for 24 h. Cell viability was then measured, and data were shown as mean ± SD from three independent experiments. Combination index (Fa-CI) plots were generated using CompuSyn Software.

safe natural sources should be considered in the future.

Cancer treatment with one single drug may not achieve a satisfactory outcome owing to primary and acquired resistance, thus providing a rationale for the combination therapy. Accumulating evidences have suggested that wogonin may serve as a suitable adjuvant to strengthen therapeutic efficacy when in combination with another agent. For instance, wogonin sensitizes malignant T cells to tumor necrosis factor-α-induced apoptosis *via* NF-κB suppression (Fas et al., 2006). Combined treatment with wogonin and tumor necrosis factor–related apoptosis-inducing ligand produces a significantly enhanced antitumor activity in a human prostate cancer cell line LNCaP *in vitro* (Lee et al., 2009) and in a non-small cell lung cancer line A549 xenografted animal model *in vivo* (Yang et al., 2013). In gastric cancer, wogonin is capable of amplifying the chemotherapeutic effect of oxaliplatin (Hong et al., 2018). Furthermore, synergistic interactions are also observed between wogonin and other agents including artesunate, oridonin, and etoposide (Chen et al., 2011; Enomoto et al., 2011; Chen et al., 2021). BCL-2 overexpression occurs in approximately 30 % of DLBCL cases, not only making it an attractive therapeutic target, but also facilitating the development of a batch of BCL-2 inhibitors (Davids, 2017). Polier *et al.* (Polier et al., 2015) suggest that wogonin potentiates the cytotoxicity of navitoclax (ABT-263) against a broad range of cancer types including

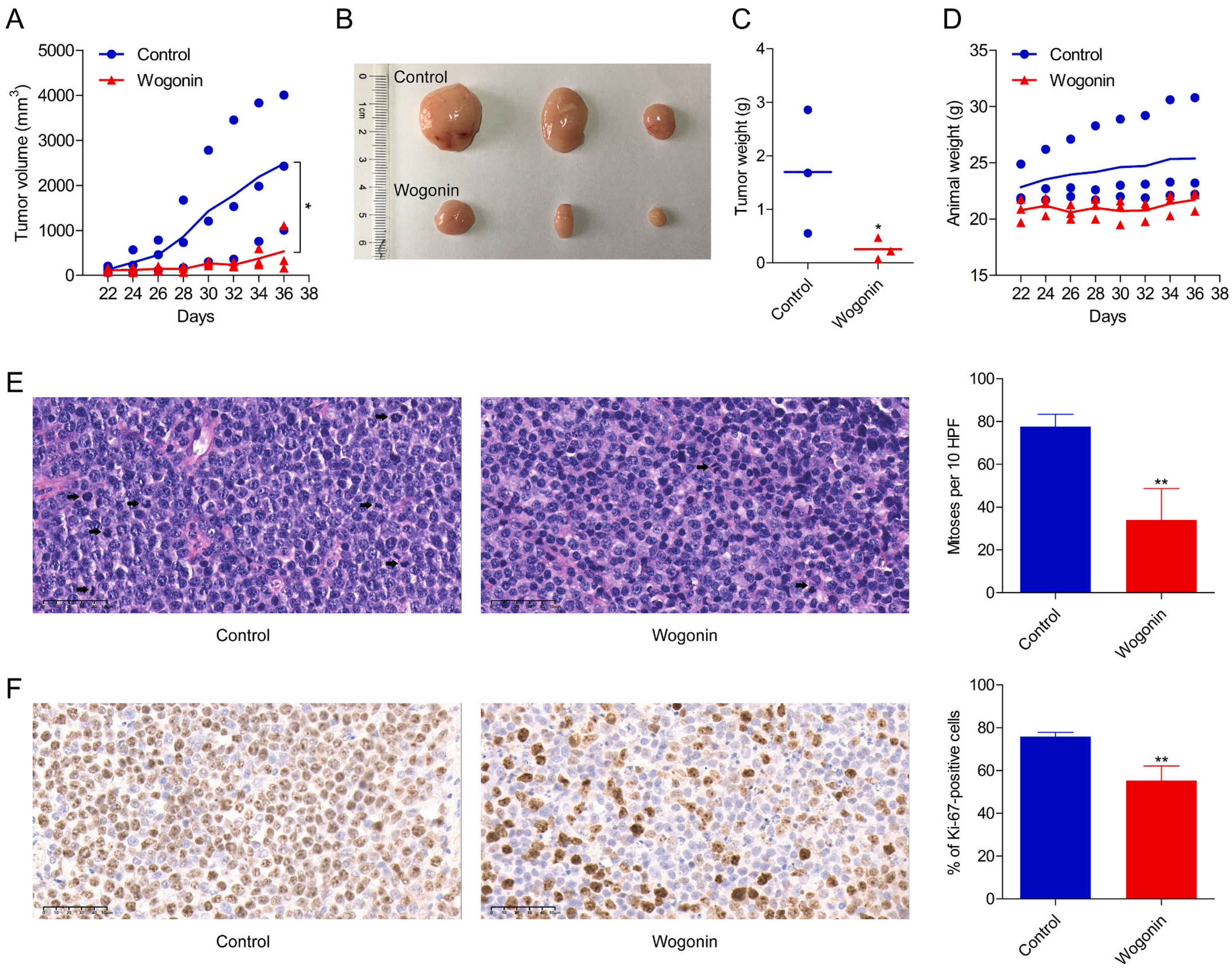

Fig. 5. Wogonin inhibits DLBCL tumor growth in a xenograft animal model *in vivo*.
(A) BALB/c-Nude mice were subcutaneously inoculated with 1×10^6 SUDHL-4 cells. Once a lump formed, mice were randomly divided into two groups ($n = 3$ for each group) and were treated with vehicle or wogonin at 25 mg/kg. The tumor volume was recorded at indicated days. *$p < 0.05$. Cohen's d: 1.74. (B—C) Tumors were photographed and weighed after removing from each mouse at the end of the experiment. *$p < 0.05$ *versus* the control group. (D) The animal body weight was monitored and measured at indicated days. (E) Representative H&E images were shown and mitoses were counted. **$p < 0.01$ *versus* the control group. (F) Representative images of immunohistochemical staining of Ki-67 on tumor tissues were shown and the percentage of Ki-67-positive cells was calculated. **$p < 0.01$ *versus* the control group.

both solid and hematological tumors, except for the human pancreatic cancer cell line MiaPaca and Hodgkin lymphoma cell line KM-H2, highlighting the potential of wogonin to promote the activity of BCL-2 inhibitors. Considering the thrombocytopenia caused by navitoclax, venetoclax (ABT-199), a more specific inhibitor of BCL-2 without targeting BCL-xL, is developed and has been approved for CLL treatment. Unfortunately, venetoclax monotherapy shows very limited efficacy in DLBCL patients, with only 18 % of overall response rate in a phase I first-in-human trial (Davids et al., 2017). A consensus has been reached so far that resistance to venetoclax is largely caused by compensatory upregulation of MCL-1 (Kapoor et al., 2020). We here identify wogonin as a potent MCL-1 inhibitor, which synergizes with venetoclax in killing DLBCL cells regardless of their molecular subtypes. Although the precise mechanism responsible for their synergy remains to be elucidated, we offer a possible combination pattern of venetoclax plus a natural compound for the treatment of patients with lymphoid malignancies.

There are several issues necessary to be addressed in future studies. First, the precise molecular mechanism responsible for the anti-DLBCL activity of wogonin remains ambiguous and identification of specific proteins targeted by wogonin is of crucial importance. Second, based on the synergy between wogonin and venetoclax, it is reasonable to further investigate whether wogonin enhances the efficacy of other targeted agents such as FLT3 inhibitors or routine chemotherapies like R-CHOP. Third, the poor solubility of wogonin always limits its pharmacological impact and optimizing the structure as well as developing efficient delivery approaches or appropriate drug carriers can be considered. Meanwhile, pharmacokinetic experiments should be conducted to monitor bioavailability, biodistribution, and metabolic conversion of wogonin.

5. Conclusion

In summary, our results reveal that wogonin treatment eliminates DLBCL cells by inducing mitochondria-dependent apoptotic cell death *in vitro* and suppresses the growth of the DLBCL tumor in an animal model *in vivo* (Fig. 6). Wogonin also enhances the activity of the BCL-2 inhibitor venetoclax in DLBCL cells regardless of their molecular subtypes. The present study suggests a therapeutic potential of wogonin in the

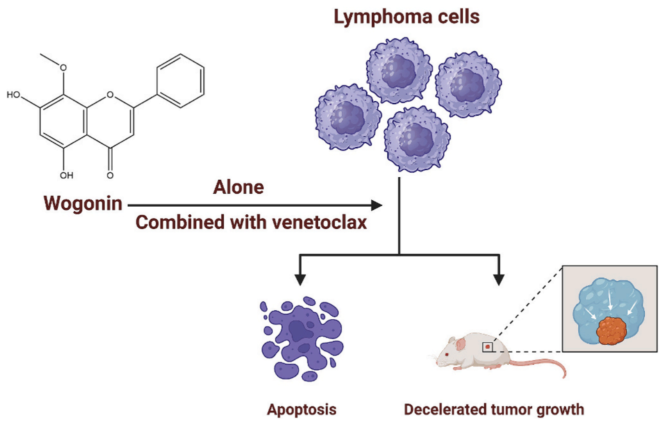

Fig. 6. Schematic illustration of the present study.
Wogonin exhibits the anti-DLBCL activity either as monotherapy or in combination with the BCL-2 inhibitor venetoclax *in vitro* and *in vivo*.

treatment of DLBCL patients and provides a basis for further evaluation in a clinical setting.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.taap.2024.117103.

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CRediT authorship contribution statement

Ye Lin: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Xia Jiang:** Methodology, Investigation. **Mengting Zhao:** Formal analysis. **Youhong Li:** Data curation. **Lili Jin:** Software. **Sumeng Xiang:** Software. **Renzhi Pei:** Writing – review & editing, Funding acquisition. **Ying Lu:** Writing – review & editing, Supervision, Funding acquisition. **Lei Jiang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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